

Sucharitha Nadendla

TECHNICAL SKILLS

Languages: Python, Java, SQL, C, Ruby

AI / ML: LLMs, Claude API (AWS Bedrock), OpenAI API, BERT Fine-tuning, PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn, Prompt Engineering

Cloud & Infra: AWS (S3, SQS, Lambda, EC2, IAM, Bedrock), Docker, Kubernetes, Terraform, Linux, Git, GitLab

Databases: MySQL, PostgreSQL, SQL Server, MongoDB

Frameworks: Flask, Django, React, Angular, Redux, REST APIs, HTML/CSS

Certifications: AWS Certified Cloud Practitioner · Oracle Cloud Data Management Associate · Oracle Cloud Infrastructure Associate

WORK EXPERIENCE

Software Engineer (AI/ML) — NCICS — Raleigh, NC

Jan 2025 – Present

- Built an end-to-end LLM-powered classification pipeline using Claude API via AWS Bedrock to label 5,560 unlabeled patient mental-health narratives, enabling downstream epidemiological analysis previously blocked by lack of labeled data.
- Fine-tuned a BERT model on Claude-generated labels, achieving production-quality classification accuracy and reducing reliance on manual expert annotation.
- Designed and deployed RESTful API endpoints in Python to expose ML and LLM-powered services to downstream research applications.
- Architected AWS data pipelines (S3, SQS, Lambda) to automate ingestion and transformation of large research datasets, improving reliability and access speed.
- Proactively proposed and led the LLM labeling initiative beyond original project scope, collaborating with two epidemiologists to align model outputs with clinical research needs.
- Added automated unit tests (Pytest) and stepwise validation to reduce regressions and improve pipeline stability across iterative development cycles.

Software Engineer — Capgemini Technology Services — Hyderabad, India

Mar 2022 – Jul 2023

- Designed and optimized SQL stored procedures for financial data pipelines processing 500,000+ records monthly, reducing query execution time by 40%.
- Built automated reporting dashboards in Python (Pandas, Matplotlib) for hospital management using Oracle EBS and BI Publisher, cutting manual reporting effort significantly.
- Developed data analytics pipelines that reduced time-to-hire by 30% by surfacing actionable recruitment insights to HR leadership.

Quality Engineer Intern — ValueLabs — Hyderabad, India

Dec 2021 – Mar 2022

- Designed and executed data integrity test plans for a medical CMS portal, increasing overall system stability by 25%.
- Validated end-to-end functionality and data consistency across content management workflows.

EDUCATION

Master of Computer Science

North Carolina State University, Raleigh, NC | Aug 2023 – May 2025 | GPA: 3.6 / 4.0

Relevant Courses: Algorithms, Software Engineering, Cloud Computing, AI

Bachelor of Engineering — Electronics

Institute of Aeronautical Engineering, India | Aug 2018 – May 2022 | GPA: 9.1 / 10

Relevant Courses: Data Structures & Algorithms, Database Management, Computer Networks

PROJECTS

LLM Mental Health Classification Pipeline (NCICS)

- Led a team of 3 to build a production pipeline using Claude API (AWS Bedrock) + BERT fine-tuning to classify 5,560 unlabeled clinical narratives; resolved IAM/Bedrock access blockers and delivered results ahead of schedule.

CodeBud — AI Coding Assistant

- Built a Flask + Streamlit web app integrating OpenAI GPT-3.5 API with structured prompt chains for code generation, explanation, and debugging.

BrainScan AI — Brain Tumor Detection

- Engineered a CNN model (Python, TensorFlow, Keras) achieving 95% accuracy classifying brain tumors from MRI scans.

Expertiza — Open-Source Peer Review Platform

- Integrated JWT authentication, reduced unauthorized access by 100%; revamped frontend in ReactJS + Redux + Tailwind; added AI-based peer-review recommendations improving platform insights by 20%.

URL Shortener

- Designed REST API in Go with Redis caching and PostgreSQL; built React + Redux frontend processing 300+ URLs monthly.